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1. **(Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#). ☹️. Unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)
- 1 / 1 point

A manufacturer can sell product 1 at a price of \$5 per unit and product 2 at a price of \$7 per unit. Nine units of raw material are needed to manufacture one unit of product 1, and seven units of raw material are needed to manufacture one unit of product 2. A total of 120 units of raw material are available. The setup costs for producing products 1 and 2 are \$30 and \$40, respectively.

To formulate an integer program that maximize the total profit, let

$$x_i = \text{sales quantity of product } i \text{ and } w_i = \begin{cases} 0 & \text{if } x_i = 0 \\ 1 & \text{if } x_i > 0 \end{cases}$$

be the decision variables. The formulation with some parts missing (labeled as A , B , and C) is

$$\begin{array}{ll} A & 5x_1 - 30w_1 + 7x_2 - 40w_2 \\ \text{s.t.} & 9x_1 + 7x_2 \leq 120 \\ & x_1 \leq Bw_1 \\ & x_2 \leq \frac{120}{9}w_2 \\ & x_i \geq 0 \quad \forall i = 1, 2, \\ & w_i \in C \quad \forall i = 1, 2. \end{array}$$

- ☐ $A = \min, B = \frac{120}{9}$, and $C = \{0, 1\}$.
- ☐ $A = \max, B = \frac{120}{7}$, and $C = \{0, 1\}$.
- ☐ $A = \max, B = \frac{120}{9}$, and $C = [0, 1]$.
- ☐ $A = \min, B = \frac{120}{7}$, and $C = [0, 1]$.
- ☒ None of the above.

👉 Correct

2. Following from the above problem, suppose that now when both products are produced for positive amounts, there is a saving of \$20 in the setup cost (i.e., in total only \$50 must be paid to set up the production processes). If the objective function is modified to
- 1 / 1 point

$$\max 5x_1 - 30w_1 + 7x_2 - 40w_2 + 20z,$$

where z is a newly added decision variable, which of the following constraints should be added to make the formulation work? Check all correct answers.

☒ $z \leq w_1$ and $z \leq w_2$.

👉 Correct

☐ $z \leq w_1 + w_2$.

☒ $2z \leq w_1 + w_2$.

👉 Correct

☐ $z \geq w_1$ and $z \geq w_2$.

☐ $2z \geq w_1 + w_2$.

3. Consider the following integer program
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$$\begin{array}{ll} \max & x_1 + x_2 \\ \text{s.t.} & 2x_1 + x_2 - 6 \leq M_1z \\ & x_1 + 2x_2 - 8 \leq M_2(1 - z) \\ & x_1 \leq 10 \\ & x_2 \leq 10 \\ & x_i \geq 0 \quad \forall i = 1, 2 \\ & z \in \{0, 1\}, \end{array}$$

where M_1 and M_2 are parameters and x_1, x_2 , and z are variables. The binary variable z is to select at least one constraint to be satisfied.

To make the program work correctly, what are the minimum possible values of M_1 and M_2 that can enable z to do the "at-least-one" selection?

- ☐ $M_1 = 3$ and $M_2 = 3$.
- ☐ $M_1 = 999999$ and $M_2 = 999999$.
- ☒ $M_1 = 24$ and $M_2 = 22$.
- ☐ $M_1 = 30$ and $M_2 = 30$.
- ☐ None of the above.

👉 Correct

4. Ten jobs should be scheduled on one single machine. The processing times for these jobs are 6, 9, 3, 5, 10, 6, 3, 9, 7, and 10 (in hours). The due times for these jobs are 50, 53, 55, 56, 60, 62, 67, 68, and 70 (in hours). We want to schedule the jobs to minimize total tardiness, which is the completion time minus the due time if positive or zero otherwise.
- 1 / 1 point

To formulate the scheduling problem as an integer program, let J be the set of jobs. Moreover, let the processing time and due time of job j be p_j and d_j , respectively. The decision variables include

$$\begin{aligned} C_j &= \text{completion time of job } j, j \in J, \\ T_j &= \text{tardiness of job } j, j \in J, \text{ and} \\ z_{ij} &= \begin{cases} 1 & \text{if job } j \text{ is scheduled before job } i, i \in J, j \in J, i < j, \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Finally, let $M = \sum_{i,j \in J} p_j$ be a large enough number. An incomplete integer program that has the potential to minimize the total tardiness is

$$\begin{array}{ll} \min & \sum_{j \in J} T_j \\ \text{s.t.} & T_j \geq C_j - d_j \quad \forall j \in J \\ & C_i + p_i - C_j \leq Mz_{ij} \quad \forall i \in J, j \in J, i < j \\ & T_j \geq 0, C_j \geq 0 \quad \forall j \in J \\ & z_{ij} \in \{0, 1\} \quad \forall i \in J, j \in J, i < j. \end{array}$$

Which of the following constraint should be added into the above program to make it complete?

- ☐ $C_j + p_i - C_i \leq Mz_{ij}$ for all $i \in J, j \in J, i < j$.
- ☒ $C_j + p_i - C_i \leq M(1 - z_{ij})$ for all $i \in J, j \in J, i < j$.
- ☐ $C_j + p_i - C_i \geq Mz_{ij}$ for all $i \in J, j \in J, i < j$.
- ☐ $C_j + p_i - C_i \geq M(1 - z_{ij})$ for all $i \in J, j \in J, i < j$.
- ☐ All of the above.

👉 Correct

5. Following from the above problem, suppose that two precedence rules must be followed: Job 1 must be finished before job 5 can start, and job 5 or job 6 must be finished before job 7 can start. Which of the following constraints should be added into the above program to ensure the precedence rules are satisfied?
- 1 / 1 point

- ☐ $z_{15} = 1$ and $z_{57} + z_{67} \leq 1$.
- ☐ $z_{15} = 1$ and $z_{57} + z_{67} \geq 1$.
- ☒ $z_{15} = 0$ and $z_{57} + z_{67} \leq 1$.
- ☐ $z_{15} = 0$ and $z_{57} + z_{67} \geq 1$.
- ☐ None of the above.

👉 Correct